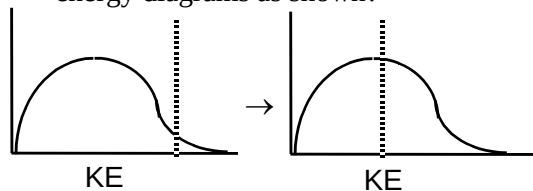


Chemical Kinetics Practice Exam

1. Which of the following does NOT influence the speed of a chemical reaction?

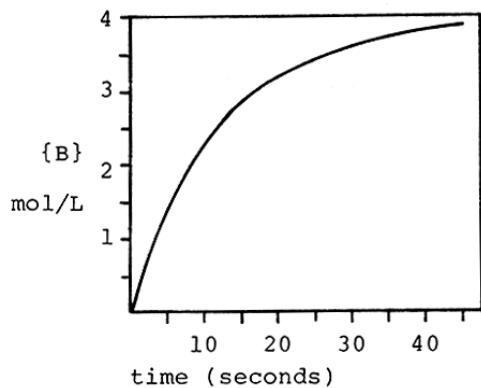
- a) concentration of reactants
- b) nature of reactants
- c) temperature
- d) presence of a catalyst
- e) none of these

2. What would cause the change in the kinetic energy diagrams as shown?



- a) increasing the ΔH
- b) decreasing the temperature
- c) increasing the surface area
- d) addition of a catalyst
- e) increasing the concentration of reactant

3. A time vs. concentration graph is presented below for the reaction $A \rightarrow B$. What is the rate of appearance of 'B' 20 seconds after the start of the reaction?



- a) 0.050 mol/L·s
- b) 3.2 mol/L·s
- c) 2.2 mol/L·s
- d) 0.010 mol/L·s
- e) 9.8 mol/L·s

4. The reaction $3O_2 \rightarrow 2O_3$ is proceeding with a rate of disappearance of O_2 equal to 0.60 mol/L·s. What is the rate of appearance of O_3 , in mol/L·s?

- a) 0.60
- b) 0.40
- c) 0.10
- d) 0.90
- e) 1.20

5. A reaction has the rate law $\text{Rate} = k[A]^2[B]$. What is the overall order of the reaction?

- a) 0
- b) 2
- c) 1
- d) 4
- e) 3

6. What are the correct units for a second order rate constant?

- a) mol/L·s
- b) 1/s
- c) L/mol·s
- d) $L^2/\text{mol}^2 \cdot s$
- e) $\text{mol}^2/L^2 \cdot s$

7. The reaction $I^- + OCl^- \rightarrow IO^- + Cl^-$ is first order with respect to I^- and first order with respect to OCl^- . The rate constant is 6.1×10^{-2} L/mol·s. What is the rate of reaction when $[I^-] = 0.10 \text{ M}$ and $[OCl^-] = 0.20 \text{ M}$?

- a) $2.4 \times 10^{-4} \text{ M/s}$
- b) $1.2 \times 10^{-3} \text{ M/s}$
- c) $6.1 \times 10^{-3} \text{ M/s}$
- d) $1.2 \times 10^{-4} \text{ M/s}$
- e) $2.4 \times 10^{-5} \text{ M/s}$

8. A reaction and its rate law are given below. When $[C_4H_6] = 2.0 \text{ M}$, the rate is 0.106 M/s . What is the rate when $[C_4H_6] = 4.0 \text{ M}$?

- $2 C_4H_6 \rightarrow C_8H_{12} \quad \text{Rate} = k[C_4H_6]^2$
- a) 0.053 M/s
 - b) 0.212 M/s
 - c) 0.106 M/s
 - d) 0.424 M/s
 - e) 0.022 M/s

9. The rate law for the reaction
 $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g})$
 is $\text{Rate} = k[\text{NO}]^2[\text{O}_2]$. What happens to the rate
 when the concentration of NO is doubled?
 a) the rate doubles d) the rate is halved
 b) the rate triples e) none of these
 c) the rate quadruples

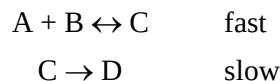
10. Below is some rate data for the hypothetical
 reaction, $2\text{A} + \text{B} \rightarrow \text{C}$. What is the rate law
 for this reaction?

Experiment	[A] ₀	[B] ₀	Rate (M/s)
1	2.0 M	1.0 M	0.100
2	2.0 M	2.0 M	0.400
3	4.0 M	1.0 M	0.100

- a) $\text{Rate} = k[\text{A}][\text{B}]$ d) $\text{Rate} = k[\text{A}]^2[\text{B}]^2$
 b) $\text{Rate} = k[\text{A}]^2[\text{B}]$ e) $\text{Rate} = k[\text{B}]^2$
 c) $\text{Rate} = k[\text{A}][\text{B}]^2$
11. The decomposition of hydrogen peroxide is a
 first order reaction with the rate constant
 given below. For an experiment in which the
 starting concentration of hydrogen peroxide
 is 0.110 M, what is the concentration of
 H_2O_2 450 minutes after the reaction begins?
 (**5 point question**)
 $2\text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{O}_2$ $k = 1.33 \times 10^{-4} \text{ min}^{-1}$
 a) 0.0961 M d) 0.00658 M
 b) 0.104 M e) 0.0156 M
 c) 0.117 M
12. What is the rate constant for a first order
 reaction for which the half-life is 85.0 sec?

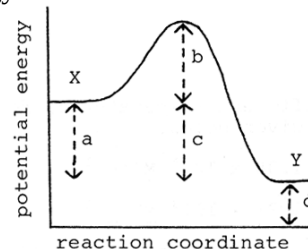
13. What fraction of a reactant remains after 3 half-
 lives of a first order reaction?
 a) 1/2 d) 1/8
 b) 1/3 e) 1/12
 c) 1/6

14. Assume a reaction occurs by the mechanism
 given below. What is the rate law for the
 reaction?



- a) $\text{Rate} = k[\text{A}][\text{B}][\text{C}]$
 b) $\text{Rate} = k[\text{A}]^2$
 c) $\text{Rate} = k[\text{A}][\text{B}]$
 d) $\text{Rate} = k[\text{A}][\text{B}]/[\text{D}]$
 e) $\text{Rate} = k[\text{A}]$
15. According to collision theory, which of the
 following factors does NOT influence the rate
 of reaction?
 a) collision frequency
 b) collision energy
 c) collision orientation
 d) collision rebound direction
 e) none of these

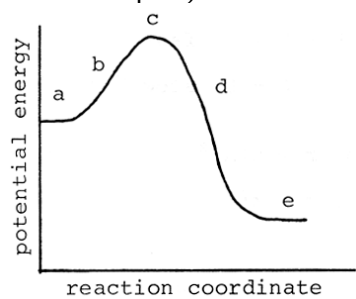
16. What distance corresponds to the activation
 energy for the reaction of X to Y?



- a) a d) d
 b) b e) e
 c) c

17. At what point on the potential energy diagram given below does the transition state (activated complex) occur?

- a) a d) d
b) b e) e
c) c



Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
E	D	A	B	E	C	B	D	C	E	B	A	D	C	D	B	C